Cardiovascular System (CARD) · NPLEX-P1-CARD-03

CV-3 Heart, Great Vessels, and Pericardium

Work through each activity on blank paper with the notes chapter closed, then open the answer key and mark what you missed. Do the two brain dumps for a competency on different days rather than back to back.

Work each activity without your notes first. Rate your confidence before you check. Then read the notes, write down what you missed, and set a date to come back to it.

Competency tracker

Competency	BD1 done	BD2 done	Confidence 1 to 4	Next review date
NPLEX-P1-CARD-03-01-C01				
NPLEX-P1-CARD-03-01-C02				
NPLEX-P1-CARD-03-01-C03				
NPLEX-P1-CARD-03-01-C04				
NPLEX-P1-CARD-03-01-C05				
NPLEX-P1-CARD-03-01-C06				

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-03-01-C01-BD1

Draw the heart in the chest and label what forms each edge

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-3 Heart, Great Vessels, and Pericardium
Competency: NPLEX-P1-CARD-03-01-C01 Activity: NPLEX-P1-CARD-03-01-C01-BD1 Date: _____

Without looking at your notes, draw the front of the chest with the sternum, the costal cartilages and the intercostal spaces, then draw the outline of the heart on it. Mark the four corners of that outline by naming the exact cartilage or intercostal space each one sits at, and mark the apex with the space it lies behind, its relation to the midclavicular line and its distance from the midline. Then write one line for each of these and name the chamber or chambers that form it: anterior surface, inferior surface, base, apex, right border, left border, inferior border. Add which compartment of the mediastinum holds the heart and which plane divides the mediastinum into superior and inferior parts. Finish by writing how the heart is rotated in the chest, how much of it lies left of the median plane, and which chamber therefore faces forward.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-03-01-C01-BD2

Read the chest film and the apex beat

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-3 Heart, Great Vessels, and Pericardium
Competency: NPLEX-P1-CARD-03-01-C01 Activity: NPLEX-P1-CARD-03-01-C01-BD2 Date: _____

Draw a posteroanterior chest film outline and label the structures that form the right contour from top to bottom and the left contour from top to bottom. Write why two of the four chambers form no edge on this view and which view shows them instead. Next, write how the cardiothoracic ratio is measured and the value above which it is abnormal. A woman with long standing mitral regurgitation has an apex beat in the sixth intercostal space lateral to the midclavicular line and a cardiothoracic ratio of 0.58 on a posteroanterior film taken in full inspiration. Write which chamber has enlarged, which border and which contour it has changed, and why the apex beat has moved. Then a man has a ratio of 0.56 on a portable anteroposterior film taken during a shallow breath. Write the two reasons that film can overstate the size of his heart and what you would ask for before calling the heart enlarged.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-03-01-C02-BD1

Lay out the chambers from front to back with their neighbors

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-3 Heart, Great Vessels, and Pericardium
Competency: NPLEX-P1-CARD-03-01-C02 Activity: NPLEX-P1-CARD-03-01-C02-BD1 Date: _____

Draw the chest in cross section at the level of the heart and lay the four chambers out in their front to back order, writing the order out in words underneath. Mark what lies immediately in front of the most anterior chamber and name the bony and cartilaginous structures it sits behind. Mark what lies immediately behind the most posterior chamber, then add the vessel behind and to its left and the two airway structures just above it. Write one line saying what separates that chamber from the esophagus. Finish by naming the embryological event that put the right ventricle at the front and the left atrium at the back.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-03-01-C02-BD2

Trace each finding of left atrial enlargement to a neighbor

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-3 Heart, Great Vessels, and Pericardium
Competency: NPLEX-P1-CARD-03-01-C02 Activity: NPLEX-P1-CARD-03-01-C02-BD2 Date: _____

A 42-year-old woman with past rheumatic fever has breathlessness on exertion, difficulty swallowing solid food and a hoarse voice. Her chest film shows a straight left heart border, a second curved edge inside the right heart border and a raised left main bronchus. For each of those six findings write the structure involved, why the left atrium reaches it rather than some other structure, and the direction the atrium had to expand to reach it. Write why the atrium cannot enlarge forwards. Then write which valve lesion her history points to and trace in three steps how that lesion raises left atrial pressure and then causes breathlessness. Finish by writing why a probe placed in the esophagus sees the left atrium and its auricle better than a probe on the chest wall, and what is most often looked for there in a patient with atrial fibrillation.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-03-01-C02-RT01

Rebuild the chain from a narrowed mitral valve to hoarseness

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-3 Heart, Great Vessels, and Pericardium
Competency: NPLEX-P1-CARD-03-01-C02 Activity: NPLEX-P1-CARD-03-01-C02-RT01 Date: _____

Rebuild the six step sequence that runs from a narrowed mitral valve to a hoarse voice, one step per line and in order. Beside the step where the atrium dilates, write the two directions it can expand and the one direction it cannot, with the reason. Beside the esophagus step, write the symptom it produces and what the barium filled esophagus looks like. Beside the bronchus step, write what happens to the angle at the carina. Beside the final step, name the nerve, the structure it hooks under and the second vessel whose dilation helps stretch it. Underneath, write the two findings this same enlargement produces on a posteroanterior film and the rhythm and complication that follow from slow flow in the dilated atrium.

Rebuild from memory: How mitral stenosis produces symptoms outside the heart

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-03-01-C03-BD1

Build the pericardium layer by layer

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-3 Heart, Great Vessels, and Pericardium
Competency: NPLEX-P1-CARD-03-01-C03 Activity: NPLEX-P1-CARD-03-01-C03-BD1 Date: _____

Draw the heart in the pericardium in a side view and label the fibrous pericardium, the parietal serous layer, the pericardial cavity and the visceral serous layer. Beside the fibrous layer write what it is made of, the three places it is attached or blended, and two consequences of the fact that it does not stretch over short periods. Beside the cavity write the normal volume of fluid in it and what that fluid does. Write one or two sentences explaining why the heart is described as lying outside the serous cavity even though the sac surrounds it. Then name the two sleeves in which the serous layer reflects, naming the vessels in each, and mark on your drawing where each of the two sinuses lies. Finish by naming the artery that supplies most of the pericardium and the artery that supplies the epicardium.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-03-01-C03-BD2

Separate pericardial pain from ischemic pain

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-3 Heart, Great Vessels, and Pericardium
Competency: NPLEX-P1-CARD-03-01-C03 Activity: NPLEX-P1-CARD-03-01-C03-BD2 Date: _____

A 34-year-old man has sharp chest pain that is worse when he breathes in deeply and when he lies flat, eases when he sits forward, and is felt along the ridge of his left trapezius. Write the nerve that carries this pain, the spinal segments it enters, which pericardial layers are pain sensitive and which are not, and the nerves that bring shoulder skin into those same segments. Then explain in three or four sentences why the pain is felt at the shoulder when nothing connects the pericardium to the shoulder. Next, write the nerves and the spinal segments that carry pain from ischemic myocardium, where that pain is felt and where it spreads. Build a short side by side list comparing the two pains on character, site, spread, effect of breathing and effect of posture. Finish by writing the added sound heard when inflamed serous surfaces rub, the number of components it can have and the cardiac events those components match.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-03-01-C03-RT01

Rebuild the referral of pericardial pain to the shoulder

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-3 Heart, Great Vessels, and Pericardium
Competency: NPLEX-P1-CARD-03-01-C03 Activity: NPLEX-P1-CARD-03-01-C03-RT01 Date: _____

Rebuild the six step sequence that runs from inflammation of the parietal pericardium to pain felt at the trapezius ridge, one step per line and in order. Beside the nerve step write the nerve, the spinal segments it enters and where it runs in relation to the fibrous pericardium, the mediastinal pleura and the root of the lung. Beside the step where the signals meet, write the nerves that bring the shoulder skin to those segments. Underneath, write in one sentence what referral depends on, then write what else irritating the diaphragm would be felt in the same place and why.

Rebuild from memory: Referral of pericardial pain to the shoulder

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-03-01-C03-CP01

Transverse against oblique pericardial sinus

Fill in every cell of the table from memory, then open the notes chapter and correct any cell you missed.

Feature	Transverse sinus	Oblique sinus
Where it lies, named by the structures in front of and behind it		
Which reflection forms each boundary		
Shape, and whether it is open or blind		
How it formed in the embryo		
Why it matters in surgery or in a patient		
Which structure just outside the pericardium lies closest to it		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-03-01-C04-BD1

Build the great vessel table from memory

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-3 Heart, Great Vessels, and Pericardium
Competency: NPLEX-P1-CARD-03-01-C04 Activity: NPLEX-P1-CARD-03-01-C04-BD1 Date: _____

On blank paper build a table with one row for each of these: superior vena cava, inferior vena cava, pulmonary veins, pulmonary trunk, ascending aorta, aortic arch. Use the columns where it begins, where it ends and its most important relation, and fill every cell. Add the three branches of the arch in their order from right to left, and write which vein crosses in front of them. Then write which of these vessels lie inside the pericardium and which do not. Underneath, draw the arch from the left side and label the ligamentum arteriosum with the two points it joins, the aortic isthmus with the two landmarks that bound it, and the aortopulmonary window. Finish by writing what the ligamentum arteriosum was in fetal life, which arch artery it came from and what that structure carried.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-03-01-C04-BD2

Three patients with vessels pressing on neighbors

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-3 Heart, Great Vessels, and Pericardium
Competency: NPLEX-P1-CARD-03-01-C04 Activity: NPLEX-P1-CARD-03-01-C04-BD2 Date: _____

Patient one is a 68-year-old smoker with a hoarse, breathy voice and a mass sitting between the underside of the aortic arch and the left pulmonary artery. Patient two has swelling of the face, neck and both arms with distended veins over the chest wall and mediastinal nodes on imaging. Patient three has difficulty swallowing and an artery that arises last from the arch and runs behind the esophagus. For each patient write the vessel or space involved, the structure being compressed or stretched, and why that structure and not another one is affected in that position. For patient one, name the nerve, the muscles it supplies and the one laryngeal muscle it does not supply, and write why a mass in the same position on the right would be far less likely to change the voice. For patient two, write the route the blood takes once the vessel is blocked. For patient three, name the variation and write what is different about the course of the laryngeal nerve on that side.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-03-01-C04-RT01

Rebuild why the two recurrent laryngeal nerves loop at different levels

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-3 Heart, Great Vessels, and Pericardium
Competency: NPLEX-P1-CARD-03-01-C04 Activity: NPLEX-P1-CARD-03-01-C04-RT01 Date: _____

Rebuild the six step sequence that explains why the two recurrent laryngeal nerves end up hooked at different levels, one step per line and in order. Beside the first step name the arch arteries both nerves start under. Beside the right sided steps write which arch arteries regress and which artery the right nerve ends up under, naming its arch of origin. Beside the left sided steps write what the left sixth arch artery becomes before birth and after birth, and the structure the left nerve stays hooked under. Then draw the arch from the left with both nerves on it, mark where each one hooks, and mark the groove each one ascends in. Finish by listing four diseases or procedures that can injure the left nerve in the chest and write in one line why its course makes it vulnerable.

Rebuild from memory: Why the two recurrent laryngeal nerves loop at different levels

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-03-01-C05-BD1

Build the four chambers from the inside

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-3 Heart, Great Vessels, and Pericardium
Competency: NPLEX-P1-CARD-03-01-C05 Activity: NPLEX-P1-CARD-03-01-C05-BD1 Date: _____

On blank paper build a table with one row for each chamber and the columns smooth part, rough part and distinctive internal features, and fill every cell. Then write one line for each of these saying exactly where it sits and what it marks or does: crista terminalis, fossa ovalis with its floor and its rim, opening of the coronary sinus, moderator band, supraventricular crest, aortic vestibule. Write how many papillary muscles each ventricle has, where their chordae attach in each ventricle, and what the papillary muscles do during systole. Add which conduction structures sit at the upper end of the crista terminalis and just above the coronary sinus opening, and which conduction structure the moderator band carries. Finish by writing in one or two sentences why atrial walls are smooth in some places and rough in others, and why the left atrium is almost entirely smooth with four openings.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-03-01-C05-BD2

Wall thickness, wall stress and two loaded ventricles

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-3 Heart, Great Vessels, and Pericardium
Competency: NPLEX-P1-CARD-03-01-C05 Activity: NPLEX-P1-CARD-03-01-C05-BD2 Date: _____

Write the normal wall thickness of each ventricle, the peak systolic pressure each one generates and the relationship between the two. Write the law of Laplace as an equation with each term named, and write what wall stress largely determines in the myocardium. Then take two patients. The first has had high aortic pressure for years. The second has had a large valve leak for years. For each one write whether the load is pressure or volume, whether sarcomeres are added in parallel or in series, what happens to wall thickness and to cavity radius, and what happens to wall stress, showing which term of the equation drives your answer. Next, write why both ventricles are the same thickness at birth, what changes afterwards and why, then explain why a large pulmonary embolism can fail the right ventricle while the same ventricle copes easily with a big volume load. Finish with the patient who has a leg vein clot, a probe patent foramen ovale and a pulmonary embolism, and an hour later develops weakness on one side: write in four steps how a clot from the leg reached a brain artery and what holds that opening shut in normal life.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Prediction exercise NPLEX-P1-CARD-03-01-C05-PR01

Predict what happens when right heart pressure rises above left

A 56-year-old woman with a probe patent foramen ovale and a deep vein thrombosis in the right calf has a large pulmonary embolism. Predict each item below, write the direction of the change, and give the structure, pressure or law that explains it in one line.

What changes	Up, down, or no change	Why, in one or two lines
What happens to pulmonary vascular resistance and to right ventricular systolic pressure		
What happens to right ventricular wall stress, naming the term of the law of Laplace that changes		
What happens to the pressure gradient across the fossa ovalis and to the flap of septum primum		
What happens to the direction of any flow through the fossa ovalis		
What happens to a fragment of thrombus arriving in the right atrium, and which filter it now escapes		
What happens to the position of the interventricular septum and to left ventricular filling		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-03-01-C06-BD1

Blunt and penetrating injury, structure by structure

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-3 Heart, Great Vessels, and Pericardium
Competency: NPLEX-P1-CARD-03-01-C06 Activity: NPLEX-P1-CARD-03-01-C06-BD1 Date: _____

Write in one sentence each how blunt trauma injures and how penetrating trauma injures, and say which structures each one damages most. Then draw the arch and descending aorta and mark the site where deceleration tears it, naming the segment and the two landmarks that bound it. Beside the drawing write why the tear happens there, naming what holds the descending aorta and what leaves the arch mobile, and write which layer can contain the leak and what that contained leak is called. List the chest film findings of this injury and the test the diagnosis actually rests on. Next write which chamber a sternal blow bruises most often, the marker it releases, the commonest rhythm disturbance and the conduction block that fits the anterior course of the right bundle. Write what a precordial blow landing on the upstroke of the T wave can trigger and the name for that. Finish by writing the order of chambers injured by stab wounds, and what decides whether blood collects in the pericardial sac or drains into the pleural cavity.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-03-01-C06-BD2

Explain every sign of tamponade from a stiff sac

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-3 Heart, Great Vessels, and Pericardium
Competency: NPLEX-P1-CARD-03-01-C06 Activity: NPLEX-P1-CARD-03-01-C06-BD2 Date: _____

A man is stabbed just left of the sternum. His pulse is 128 per minute, his blood pressure is 82 over 64 mm Hg, his neck veins are distended and his heart sounds are muffled. Breath sounds are equal on both sides and the trachea is midline. Write which chamber the blade most likely entered and why that chamber and not another. Then write how much blood in the sac can cause this when it enters over minutes, and explain from the shape of the pericardial pressure and volume curve why a slowly collecting effusion can reach a liter with little pressure rise. Explain each of the following in one or two sentences from rising pressure in a stiff closed sac: why the right chambers are affected before the left, why filling and stroke volume fall, why venous pressure rises, why diastolic pressures in all four chambers become equal, why the heart sounds are muffled and why the pulse is fast. Then write why systolic pressure falls by more than 10 mm Hg on inspiration, naming what the septum does and why it has nowhere else to go. Write what happens to the jugular y descent, and write why the neck veins can be flat in a patient who has also lost a lot of blood. Finish by writing the two findings on examination that would point to a tension pneumothorax instead.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-03-01-C06-RT01

Rebuild the downward spiral of acute tamponade

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-3 Heart, Great Vessels, and Pericardium
Competency: NPLEX-P1-CARD-03-01-C06 Activity: NPLEX-P1-CARD-03-01-C06-RT01 Date: _____

Rebuild the six step loop of acute tamponade, one step per line and in order, then draw it as a closed circle with arrows so each step feeds the next. Beside the first step write the volume of rapidly entering blood that can do this and the property of the fibrous sac that allows it. Beside the compression step write which chambers are squeezed first and in which part of the cardiac cycle. Beside the coronary step write the two pressures whose difference gives coronary perfusion pressure and what each one is doing here. Underneath the loop write the rhythm the spiral ends in if the blood is not removed, and write in one line what breaks the loop.

Rebuild from memory: The downward spiral of acute tamponade

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-03-01-C06-CP01

Tamponade against tension pneumothorax after chest trauma

Fill in every cell of the table from memory, then open the notes chapter and correct any cell you missed.

Finding	Cardiac tamponade	Tension pneumothorax
Why cardiac output falls		
Neck veins		
Breath sounds and percussion note		
Position of the trachea		
Heart sounds		
What makes the neck veins flat in either condition		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Clinical application NPLEX-P1-CARD-03-01-C06-CA01

Shock with distended neck veins after a stab wound

A 27-year-old man is brought to the emergency department 20 minutes after being stabbed with a short blade. The wound is in the fourth intercostal space 2 cm to the left of the sternal edge. Temperature is 36.8 °C (98.2 °F), pulse is 132 per minute, respirations are 26 per minute and blood pressure is 80 over 66 mm Hg. The neck veins are distended at 45 degrees, the heart sounds are muffled and the precordium is quiet. Breath sounds are equal on both sides and the percussion note is normal in both lungs. The trachea is midline. Systolic pressure falls by 18 mm Hg during quiet inspiration. The jugular pulse shows a brisk inward movement with the x descent and almost no y descent. The electrocardiogram shows sinus tachycardia with normal voltages and no alternans.

1. Recognize. Name the chamber the blade most likely entered and the condition that has developed in the pericardial sac, and name the three findings in this patient that together make up the classic triad.

2. Explain. Explain why a volume of blood this small produces shock, and explain why filling of the right chambers fails before the left.

3. Differentiate. Write how this picture differs from a tension pneumothorax and from blood loss alone, naming the findings in the vignette that separate them.

4. Investigate. Write which measurements and which findings on the chest film and on bedside imaging you would look for, and what each one would show in this patient.

5. Decide. State which chamber pressures you expect to be equal in diastole, and state what the inspiratory fall in systolic pressure and the blunted y descent each tell you about filling.

6. Justify. Justify your answer from the structure of the pericardium and from the way the two ventricles share a fixed space, and explain why a slow effusion of the same volume would not do this.

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-03-01-C01-Q01

Which chamber forms the apex beat

A 54-year-old woman is examined because of 6 months of breathlessness on exertion. A forceful, sustained impulse is felt in the left sixth intercostal space 3 cm lateral to the midclavicular line. A posteroanterior chest radiograph taken in full inspiration shows a cardiothoracic ratio of 0.59 with the lower left contour displaced downward and to the left. Enlargement of which of the following chambers best accounts for these findings?

A. Right atrium

B. Right ventricle

C. Left atrium

D. Left ventricle

Why did you pick it?

Board-style question NPLEX-P1-CARD-03-01-C02-Q02

Dysphagia and a widened carina

A 45-year-old woman with a history of rheumatic fever has 4 months of difficulty swallowing solid food. A barium swallow shows an indentation on the anterior wall of the esophagus at the level of the carina, and a chest radiograph shows elevation of the left main bronchus with a widened carinal angle. Which of the following best explains these findings?

A. An enlarged left atrium expanding backward and upward against the esophagus and left main bronchus

B. A dilated right ventricle filling the retrosternal space and displacing the mediastinum

C. An aneurysm of the descending thoracic aorta indenting the esophagus from behind

D. An aberrant right subclavian artery passing behind the esophagus

Why did you pick it?

Board-style question NPLEX-P1-CARD-03-01-C04-Q03

Hoarseness from a mediastinal mass

A 68-year-old man who has smoked for 40 years has 2 months of a hoarse, breathy voice. Computed tomography shows a 4 cm mass in the space bounded above by the inferior surface of the aortic arch and below by the left pulmonary artery. Injury to a nerve at which of the following sites best explains his voice change?

A. The right recurrent laryngeal nerve where it hooks under the right subclavian artery

B. The left phrenic nerve where it crosses the aortic arch

C. The left recurrent laryngeal nerve where it hooks under the arch at the ligamentum arteriosum

D. The left vagus nerve where it lies behind the root of the left lung

Why did you pick it?

Board-style question NPLEX-P1-CARD-03-01-C06-Q04

Where deceleration tears the aorta

A 31-year-old man is brought to the emergency department after a head-on collision at high speed. He is hypotensive. A chest radiograph shows a widened mediastinum, loss of the normal aortic knob contour, deviation of the trachea and nasogastric tube to the right, and fractures of the left first and second ribs. Computed tomographic angiography shows an aortic injury. At which of the following sites is the tear most likely to be found, and for which reason?

A. The ascending aorta, because it lies inside the pericardium and is compressed against the sternum

B. The aortic isthmus, because the mobile arch shears against the descending aorta held to the spine

C. The aortic root at the valve, because the aortic and mitral valves are in fibrous continuity

D. The aorta at the diaphragmatic hiatus, because the aortic opening is the narrowest point

Why did you pick it?